The Complete Treatise on the Analysis of a Duopoly

An Application to Norway

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This treatise develops a general theory of duopoly as a system of two strategically interacting entities. It rejects the restriction that the entities must be firms: they may be sovereign states, central banks, currencies, banks, exchanges, technological platforms, alliances, public institutions, or other identifiable agents. The theory combines set theory, game theory, optimization, dynamic systems, network analysis, financial economics, econometrics, law, public policy, and computational methods. Norway provides the institutional and empirical application environment. The Norwegian application is deliberately modular: the two entities must be identified for each research design rather than assumed in advance. The treatise distinguishes a two-entity system from a true duopoly, formalizes reciprocal strategic interdependence, derives benchmark equilibria, develops dynamic and stochastic extensions, proposes econometric identification strategies, and specifies a reproducible computational research programme. Norwegian institutional facts are separated from theoretical assumptions and empirical hypotheses. No unestimated coefficient, causal effect, or empirical finding is presented as established.

The treatise ends with "The End"

Keywords: duopoly; strategic interdependence; Norway; industrial organization; game theory; sovereign economics; monetary institutions; financial networks; competition law; econometrics; systemic risk; dynamic games.

Contents

1 Introduction **4**

1.1 Contributions 5

1.2 Scope and epistemic discipline 5

2 Ontology and Definition **5**

2.1 Entities and analytical domains 5

2.2 A general system representation 6

2.3 Axioms 6

3 Taxonomy of Duopolies **7**
3.1 Institutional identity . . . 7
3.2 Homogeneity and heterogeneity . . . 7
3.3 Symmetry and asymmetry . . . 8
3.4 Multidimensional classification . . . 8

4 Set Theory, Action Spaces, and Strategic Games **8**
4.1 Feasible actions . . . 8
4.2 Objective functions . . . 9
4.3 Best responses and Nash equilibrium . . . 9
4.4 Strategic interdependence . . . 9

5 Interaction Matrices and Network Representation **9**
5.1 Local influence matrix . . . 9
5.2 Directed weighted graph . . . 10
5.3 Nested systems . . . 10

6 Classical Economic Benchmarks **10**
6.1 Cournot quantity competition . . . 11
6.2 Bertrand price competition . . . 11
6.3 Stackelberg leadership . . . 11
6.4 Generalized non-firm interpretation . . . 12

7 Existence, Uniqueness, and Stability **12**
7.1 Existence . . . 12
7.2 Uniqueness . . . 12
7.3 Dynamic stability . . . 12

8 Dynamic Games, Learning, and Evolution **13**
8.1 Intertemporal objective . . . 13
8.2 State transition . . . 13
8.3 Repeated interaction . . . 13
8.4 Evolutionary and learning dynamics . . . 13

9 Resources, Capital, and Financial Planning **14**
9.1 Resource accumulation . . . 14
9.2 Risk-adjusted resource allocation . . . 14
9.3 Liquidity and solvency . . . 14
9.4 Financial asymmetry . . . 14

10 Information, Uncertainty, and Behaviour **14**
10.1 Information structure . . . 14
10.2 Beliefs and expectations . . . 15
10.3 Behavioural and psychological channels . . . 15
10.4 Linguistic and narrative channels . . . 15

11 Norway: Institutional and Economic Application **15**
11.1 Norway as an application environment . . . 15
11.2 Norges Bank . . . 15
11.3 Norwegian financial regulation . . . 16
11.4 Possible Norway research designs . . . 16
11.5 Norway-specific empirical discipline . . . 16

12 Law, Regulation, and Institutional Design **17**
12.1 Legal feasibility . . . 17
12.2 Competition law . . . 17
12.3 Public institutions . . . 17
12.4 Ethics and rights . . . 17

13 Welfare Economics and Distribution **18**
13.1 Social welfare . . . 18
13.2 Surplus and externalities . . . 18
13.3 Distributional incidence . . . 18

14 Geopolitics, Diplomacy, and Security Economics **18**
14.1 Warfare economics and conflict risk . . . 18

15 Physics, Chemistry, Biology, and Environmental Constraints **19**
15.1 Physics and energy . . . 19
15.2 Chemistry and materials . . . 19
15.3 Biology and public health . . . 19
15.4 Environmental accounting . . . 20

16 Econometric Framework **20**
16.1 Research estimand . . . 20
16.2 Panel model . . . 20
16.3 Simultaneity . . . 20
16.4 Time-series dynamics . . . 20
16.5 Causal designs . . . 21
16.6 Concentration measures . . . 21

17 Statistics, Uncertainty, and Robustness **21**
17.1 Sampling uncertainty . . . 21
17.2 Model uncertainty . . . 21
17.3 Stress testing . . . 21

18 Computer Science, Algorithms, and AI/ML **21**
18.1 Computational representation . . . 21
18.2 Equilibrium computation . . . 22
18.3 Illustrative pseudocode . . . 22
18.4 Machine learning . . . 23
18.5 Cybersecurity and data governance . . . 23

19 The Duopoly Transition Problem **23**
19.1 From monopoly to duopoly . . . 23
19.2 From duopoly to oligopoly . . . 23
19.3 From duopoly to monopoly . . . 24
19.4 Reversibility . . . 24

20 Propositions and Research Hypotheses **24**

21 Philosophical and Methodological Foundations **25**

22 Empirical Research Protocol for Norway **25**
22.1 Step 1: Select the dyad . . . 25
22.2 Step 2: Define the outcome . . . 25
22.3 Step 3: Map the institutional environment . . . 25
22.4 Step 4: Assemble data . . . 25
22.5 Step 5: Estimate and validate . . . 26
22.6 Step 6: Interpret . . . 26

23 Limitations **26**

24 Conclusion **26**

A Notation **27**

List of Figures

1 A reciprocal two-entity interaction network. Edge weights require domain-specific interpretation. . . . 10

List of Tables

1 Possible entity types in a general duopoly . . . 7
2 Dimensions for classifying a duopoly . . . 8
3 Potential duopoly applications involving Norway . . . 16

1 Introduction

A duopoly is commonly introduced as a market supplied by two firms. That definition is useful in industrial organization, but it is too narrow for a general theory of two-entity strategic interaction. Two sovereign states may compete over trade, investment, security, or technological standards. Two central banks may interact through exchange rates, liquidity, payment systems, and cross-border financial conditions. Two currencies may compete as stores of value or settlement media. Two exchanges, payment networks, or technological ecosystems may interact through network effects and interoperability.

The correct starting point is therefore not the legal form of the participants but the structure of their interaction. The analysis must identify (i) two distinct entities, (ii) a specified domain of interaction, (iii) feasible actions, (iv) objectives or behavioural

criteria, (v) constraints, and (vi) a mechanism through which one entity's choices affect the other's outcomes.

This generality must not be confused with the claim that every pair of entities forms a duopoly. Two entities may coexist without interacting. They may interact without influencing each other's decisions. They may be two members of a much larger market in which neither is pivotal. The term "duopoly" is appropriate only when the chosen analytical boundary makes the two entities the strategically relevant participants in the specified domain.

The Norway application is treated as an institutional setting, not as an assumption that Norway itself must be one of the two entities. Norway may be studied as a sovereign participant, as a jurisdiction containing institutions, as a market, as a monetary system, or as the environment in which two external or domestic entities interact. A particular empirical study must state which interpretation it adopts.

1.1 Contributions

The treatise makes six methodological contributions.

1. It defines duopoly in terms of strategic structure rather than corporate identity.
2. It distinguishes cardinality, interaction, strategic interdependence, market dominance, and institutional scope.
3. It integrates static games, dynamic games, stochastic processes, networks, and resource accumulation.
4. It derives benchmark results while stating the assumptions under which they hold.
5. It develops a Norway-oriented legal, institutional, and data architecture.
6. It specifies a reproducible computational and econometric programme.

1.2 Scope and epistemic discipline

The manuscript separates established results from proposed definitions, model-dependent propositions, and empirical hypotheses. A mathematical theorem is valid only under its stated assumptions. A Norwegian institutional description is not itself evidence of a duopoly. A descriptive concentration measure is not a causal explanation. A model simulation is not an empirical estimate.

2 Ontology and Definition

2.1 Entities and analytical domains

Let the two candidate entities be

$$\mathcal{A} = \{A_1, A_2\}. \tag{1}$$

The entities are distinct:

$$A_1 \neq A_2. \tag{2}$$

They may be homogeneous or heterogeneous in institutional type, legal status, resources, objectives, and decision rules.

Let Ω denote the analytical domain. Examples include a product market, a currency market, a bilateral sovereign relationship, a payment network, a technology standard, or a regulatory function. The same two entities may constitute a duopoly in one domain and not in another.

Definition 2.1 (Two-entity system)**.** A two-entity system is a specified analytical system whose focal entity set is $\mathcal{A} = \{A_1, A_2\}$, with $A_1 \neq A_2$.

Definition 2.2 (Strategic interdependence)**.** Entities A_1 and A_2 are strategically interdependent in domain Ω if at least one feasible action of each entity can alter the other entity's attainable outcomes, incentives, constraints, or optimal responses in that domain.

Definition 2.3 (Duopoly)**.** A duopoly in domain Ω is a two-entity strategic system in which A_1 and A_2 are the two strategically relevant focal entities and their decisions are mutually interdependent, or are linked by an explicitly modelled asymmetric strategic relationship. The definition is conditional on the declared analytical boundary and does not imply that no peripheral participants exist.

The reciprocal case is the principal case studied here. A one-way dependence may be analysed as a boundary case, but it should not be silently treated as reciprocal strategic competition.

2.2 A general system representation

Define

$$\mathfrak{D} = (\mathcal{A}, \Omega, \mathcal{E}, \mathcal{S}, \mathcal{C}, \mathcal{X}, \mathcal{U}, \mathcal{I}, \mathcal{T}), \tag{3}$$

where $\mathcal{E}$ is the environment, $\mathcal{S}$ the state space, $\mathcal{C}$ the constraint system, $\mathcal{X}$ the action space, $\mathcal{U}$ the objective or behavioural criteria, $\mathcal{I}$ the interaction structure, and $\mathcal{T}$ the time domain.

The representation is deliberately typed. For a commercial market, objectives may be profits and actions may be prices or quantities. For a central bank, the objective may be a legally specified policy mandate and actions may be policy instruments. For sovereign states, the model may include welfare, security, fiscal capacity, trade, and diplomatic choices. These applications share a mathematical architecture but not necessarily the same substantive objective function.

2.3 Axioms

Axiom 2.4 (Cardinality)**.** The focal set contains exactly two entities: $|\mathcal{A}| = 2$.

Axiom 2.5 (Distinguishability)**.** The entities are distinct in the chosen domain: $A_1 \neq A_2$.

Axiom 2.6 (Domain specification)**.** The analyst must state the market, function, jurisdiction, network, or strategic domain in which the duopoly is asserted.

Axiom 2.7 (Strategic relevance)**.** Both entities must be relevant to the outcomes under investigation. Mere co-presence is insufficient.

Axiom 2.8 (Heterogeneous resources)**.** The entities may possess unequal resources, capabilities, information, and institutional authority.

Axiom 2.9 (Heterogeneous objectives)**.** The entities need not maximize the same objective, and an entity's objective may be multi-dimensional or legally constrained.

Axiom 2.10 (Temporal evolution)**.** The entities, their feasible actions, and their environment may evolve over time.

Axiom 2.11 (Institutional constraint)**.** The feasible actions of each entity are restricted by applicable law, contracts, governance arrangements, technology, resources, and external conditions.

Axiom 2.12 (Cooperation and competition)**.** The entities may compete in one dimension and cooperate in another. Cooperation does not by itself eliminate their distinct identities or strategic interdependence.

Remark 2.13. The axioms are modelling commitments, not empirical laws. In particular, cardinality refers to the focal strategic entities, not necessarily every participant in the wider environment.

3 Taxonomy of Duopolies

3.1 Institutional identity

Table 1: Possible entity types in a general duopoly

Domain	Entity A_1	Entity A_2
Corporate	Firm	Firm
Sovereign	State	State
Monetary	Currency	Currency
Central banking	Central bank	Central bank
Financial	Bank	Bank
Market infrastructure	Exchange	Exchange
Technology	Platform	Platform
Geopolitical	Alliance	Alliance
Institutional	Public institution	Private institution
Hybrid	State-owned entity	Private entity

3.2 Homogeneity and heterogeneity

A homogeneous duopoly contains entities of the same broad institutional class. A heterogeneous duopoly contains entities of different classes. For example, a sovereign authority and a private financial institution may interact strategically, but their legal powers and objective functions must be modelled separately.

3.3 Symmetry and asymmetry

Let the resource vector of entity i be

$$\mathbf{R}_i = (K_i, L_i, T_i, I_i, N_i, H_i), \tag{4}$$

where K_i denotes financial resources, L_i labour and human capital, T_i technology, I_i information, N_i network reach, and H_i institutional or legal authority. A resource-asymmetric system has

$$\mathbf{R}_1 \neq \mathbf{R}_2. \tag{5}$$

This does not prove that one entity dominates. Dominance depends on how resources translate into feasible actions and outcomes.

3.4 Multidimensional classification

Table 2: Dimensions for classifying a duopoly

Dimension	Possible forms
Institutional identity	Homogeneous; heterogeneous
Resources	Symmetric; asymmetric
Strategic influence	Balanced; unbalanced
Information	Complete; incomplete; asymmetric
Timing	Simultaneous; sequential; repeated
Interaction	Competitive; cooperative; mixed
Market access	Open; restricted; regulated
Technology	Interoperable; incompatible; networked
Risk	Symmetric; asymmetric; correlated
Geography	Domestic; bilateral; international
Legal status	Public; private; mixed; supranationally constrained

These dimensions are not mutually exclusive categories. A system may be heterogeneous, resource-asymmetric, repeated, internationally regulated, and partly cooperative at the same time.

4 Set Theory, Action Spaces, and Strategic Games

4.1 Feasible actions

Let $\mathcal{X}_i$ be the feasible action set of entity A_i. The joint action space is

$$\mathcal{X} = \mathcal{X}_1 \times \mathcal{X}_2. \tag{6}$$

An action may be a price, quantity, interest rate, investment, fiscal measure, exchange-rate intervention, technology choice, security posture, regulatory decision, or diplomatic proposal.

The feasible set is not necessarily unconstrained. Write

$$\mathcal{X}_i = \{x_i \in \mathbb{R}^{n_i} : g_i(x_i, \mathbf{z}) \leq 0,\ h_i(x_i, \mathbf{z}) = 0\}, \tag{7}$$

where $\mathbf{z}$ collects relevant state variables and institutional constraints.

4.2 Objective functions

For a complete-information static game, define

$$U_1 : \mathcal{X}_1 \times \mathcal{X}_2 \to \mathbb{R}, \tag{8}$$
$$U_2 : \mathcal{X}_1 \times \mathcal{X}_2 \to \mathbb{R}. \tag{9}$$

The strategic game is

$$\mathcal{G} = (\{A_1, A_2\}, \mathcal{X}_1, \mathcal{X}_2, U_1, U_2)\,. \tag{10}$$

The scalar representation is a simplifying device. Multi-objective, lexicographic, rights-constrained, or satisficing decision rules may be more appropriate for public institutions and sovereign actors.

4.3 Best responses and Nash equilibrium

The best-response correspondence of entity i is

$$BR_i(x_j) = \arg\max_{x_i \in \mathcal{X}_i} U_i(x_i, x_j). \tag{11}$$

A Nash equilibrium is a profile (x_1^*, x_2^*) satisfying

$$x_1^* \in BR_1(x_2^*), \tag{12}$$
$$x_2^* \in BR_2(x_1^*). \tag{13}$$

Equivalently, for every feasible unilateral deviation,

$$U_1(x_1^*, x_2^*) \geq U_1(x_1, x_2^*), \tag{14}$$
$$U_2(x_1^*, x_2^*) \geq U_2(x_1^*, x_2). \tag{15}$$

The equilibrium concept does not imply social optimality, fairness, stability under learning, or legal permissibility.

4.4 Strategic interdependence

In differentiable environments, a useful local diagnostic is

$$\frac{\partial U_1}{\partial x_2} \neq 0, \qquad \frac{\partial U_2}{\partial x_1} \neq 0. \tag{16}$$

This is a sufficient indicator of direct marginal cross-effects in the chosen parameterization, not a universal definition. Strategic interdependence can also arise through constraints, beliefs, discontinuities, or discrete actions.

5 Interaction Matrices and Network Representation

5.1 Local influence matrix

Define a local influence matrix

$$\mathbf{J} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix}, \qquad J_{ij} = \frac{\partial U_i}{\partial x_j}, \tag{17}$$

where the derivative exists. Diagonal terms capture own-action marginal effects; off-diagonal terms capture cross-effects. The sign and magnitude of the cross-effects depend on the model and scaling.

A different matrix may be needed for state transmission:

$$\mathbf{M} = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix}. \tag{18}$$

It is important not to conflate an objective-gradient matrix with a causal transmission matrix. The former is a mathematical derivative; the latter requires a structural interpretation and, for empirical claims, an identification strategy.

5.2 Directed weighted graph

Let $G = (V, E, W)$, where $V = \{A_1, A_2\}$, E is a set of directed edges, and W assigns weights to the edges. A directed edge $A_j \to A_i$ represents a specified channel through which A_j affects A_i.

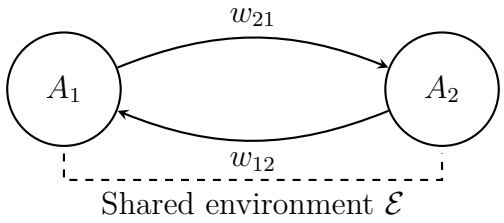


Figure 1: A reciprocal two-entity interaction network. Edge weights require domain-specific interpretation.

The graph can be multilayered. One layer may represent trade, another financial exposures, another information transmission, and another legal authority. A single edge should not combine conceptually different mechanisms unless the aggregation is justified.

5.3 Nested systems

A duopoly at one analytical level may contain many entities internally. Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be two blocs, each containing multiple members. The external system is

$$\mathcal{D}_{\text{external}} = \{\mathcal{B}_1, \mathcal{B}_2\}. \tag{19}$$

The internal structure of each bloc may be an oligopoly, coalition, federation, hierarchy, or network. This is the nested-duopoly problem. The analyst must state the aggregation rule and demonstrate that internal heterogeneity does not invalidate the chosen representation.

6 Classical Economic Benchmarks

The classical models are special cases of the general theory, not its definition. Cournot, Bertrand, and Stackelberg models are central benchmarks in industrial organization [4, 3, 17, 20, 11].

6.1 Cournot quantity competition

Assume two suppliers choose quantities $q_1, q_2 \geq 0$, aggregate output is $Q = q_1 + q_2$, inverse demand is $P(Q) = a - bQ$, and constant marginal cost is c_i, with $a > c_i$ and $b > 0$. Profit is

$$\pi_i(q_i, q_j) = (a - b(q_i + q_j) - c_i)\, q_i. \tag{20}$$

The first-order condition for an interior optimum is

$$\frac{\partial \pi_i}{\partial q_i} = a - c_i - 2bq_i - bq_j = 0. \tag{21}$$

Hence the best response is

$$BR_i(q_j) = \max\left\{0, \frac{a - c_i - bq_j}{2b}\right\}. \tag{22}$$

For symmetric costs $c_1 = c_2 = c$, the symmetric equilibrium is

$$q_1^* = q_2^* = \frac{a - c}{3b}, \qquad Q^* = \frac{2(a - c)}{3b}, \qquad P^* = \frac{a + 2c}{3}. \tag{23}$$

The second derivative is

$$\frac{\partial^2 \pi_i}{\partial q_i^2} = -2b < 0, \tag{24}$$

so the objective is strictly concave in own quantity. The first-order condition therefore identifies the unique best response for a fixed competitor quantity. The simultaneous linear best-response equations yield the stated equilibrium.

6.2 Bertrand price competition

Suppose entities choose prices. Under homogeneous products, identical constant marginal costs, no capacity constraints, and the standard undercutting assumptions, the Bertrand equilibrium price equals marginal cost:

$$p_1^* = p_2^* = c. \tag{25}$$

This result is highly assumption-dependent. Product differentiation, capacity limits, switching costs, search frictions, network effects, repeated interaction, and institutional constraints can change the equilibrium [3, 20].

6.3 Stackelberg leadership

Suppose entity 1 chooses quantity first and entity 2 observes and responds. With symmetric linear demand and costs, the follower's response is

$$q_2(q_1) = \frac{a - c - bq_1}{2b}, \tag{26}$$

when the expression is positive. The leader substitutes this response into its own objective and maximizes over q_1. The interior solution is

$$q_1^* = \frac{a - c}{2b}, \qquad q_2^* = \frac{a - c}{4b}. \tag{27}$$

The result illustrates how timing and commitment alter outcomes. It does not imply that every asymmetric duopoly has a leader-follower structure.

6.4 Generalized non-firm interpretation

The algebra of a quantity game may be reused for two sovereign exporters, two payment networks, or two institutions only if the quantities and payoff functions have meaningful domain-specific interpretations. A mathematical isomorphism does not establish substantive equivalence.

7 Existence, Uniqueness, and Stability

7.1 Existence

A standard existence result applies when each strategy set is nonempty, compact, and convex, and each payoff function is continuous in all strategies and quasi-concave in a player's own strategy. Under these conditions, a Nash equilibrium exists [12, 13, 6].

Theorem 7.1 (Existence of pure-strategy equilibrium under standard conditions)**.** *Let $\mathcal{X}_1, \mathcal{X}_2$ be nonempty, compact, and convex subsets of finite-dimensional Euclidean spaces. Suppose U_i is continuous in (x_1, x_2) and quasi-concave in x_i for each fixed x_j. Then the game has at least one pure-strategy Nash equilibrium.*

Proof. For each player, continuity and compactness ensure that the best-response correspondence is nonempty and compact-valued. Quasi-concavity makes the best-response values convex. Under the stated continuity conditions, the best-response correspondence is upper hemicontinuous. The product best-response correspondence maps the compact convex joint strategy space into itself. Kakutani's fixed-point theorem therefore supplies a fixed point. At that fixed point, each player's strategy is a best response to the other's, which is precisely a Nash equilibrium. □

7.2 Uniqueness

Existence does not imply uniqueness. Uniqueness may be established through contraction properties, diagonal strict concavity, monotonicity, or other model-specific conditions [14]. A claimed unique equilibrium must state and verify the relevant assumptions.

7.3 Dynamic stability

Let the state vector be

$$\mathbf{Z}(t) = \begin{pmatrix} \mathbf{Z}_1(t) \\ \mathbf{Z}_2(t) \\ \mathbf{Z}_E(t) \end{pmatrix}, \tag{28}$$

where $\mathbf{Z}_i$ describes entity i and $\mathbf{Z}_E$ the environment. A continuous-time system is

$$\dot{\mathbf{Z}}(t) = \mathbf{F}(\mathbf{Z}(t), \mathbf{x}(t), t). \tag{29}$$

At a stationary equilibrium,

$$\mathbf{F}(\mathbf{Z}^*, \mathbf{x}^*, t) = \mathbf{0}, \tag{30}$$

for an autonomous or appropriately stationary formulation.

The Jacobian is

$$\mathbf{J}_F = \left. \frac{\partial \mathbf{F}}{\partial \mathbf{Z}} \right|_{\mathbf{Z}=\mathbf{Z}^*}. \tag{31}$$

If all eigenvalues of $\mathbf{J}_F$ have strictly negative real parts, the equilibrium is locally asymptotically stable for the differentiable autonomous system. This is a local result; it does not establish global stability or resilience to large shocks.

8 Dynamic Games, Learning, and Evolution

8.1 Intertemporal objective

For entity i, a general discounted objective is

$$V_i(\mathbf{z}_0) = \sup_{\{x_i(t)\}} \mathbb{E}_0 \left[\int_0^\infty e^{-\rho_i t} u_i(\mathbf{z}(t), x_i(t), x_j(t))\, dt \right], \tag{32}$$

subject to state dynamics, feasibility constraints, and the strategy of the other entity. Here $\rho_i > 0$ is a discount rate. For public institutions, the discounting and objective specification require normative justification.

8.2 State transition

A general discrete-time system is

$$\mathbf{z}_{t+1} = F(\mathbf{z}_t, x_{1t}, x_{2t}, \varepsilon_{t+1}), \tag{33}$$

where ε_{t+1} is an external shock. The corresponding stochastic game requires beliefs, information sets, and strategies conditioned on available information.

8.3 Repeated interaction

If the stage game is repeated, a strategy may condition on the history $\mathcal{H}_t$. Repetition can support outcomes that are not equilibria of the one-shot game, depending on discount factors, observability, credible punishment, and information. The folk-theorem family of results requires precise assumptions and should not be interpreted as a universal prediction of cooperation [5].

8.4 Evolutionary and learning dynamics

Let $s_i(t)$ denote a strategy share or state variable. A replicator-style equation is

$$\dot{s}_k = s_k \left(f_k(\mathbf{s}) - \bar{f}(\mathbf{s}) \right), \tag{34}$$

where f_k is the payoff to strategy k and $\bar{f}$ the population average. In a strict two-entity model this is not automatically an appropriate population model; it becomes relevant when each entity contains adaptive subunits or when strategies diffuse through a wider population [19, 21].

9 Resources, Capital, and Financial Planning

9.1 Resource accumulation

Let $K_i(t)$ be a financial or productive capital stock. A simple accumulation equation is

$$\dot{K}_i(t) = I_i(t) - \delta_i K_i(t), \tag{35}$$

where I_i is investment and δ_i depreciation. A budget constraint may be written

$$I_i(t) + C_i(t) + D_i(t) \leq Y_i(t) + B_i(t), \tag{36}$$

where C_i is current expenditure, D_i debt service, Y_i internally generated resources, and B_i net financing. The terms must be adapted to the entity: a sovereign budget is not a corporate cash-flow statement.

9.2 Risk-adjusted resource allocation

A generic portfolio problem is

$$\max_{\mathbf{w}_i} \left\{ \mathbb{E}[R_i(\mathbf{w}_i)] - \frac{\gamma_i}{2} \operatorname{Var}(R_i(\mathbf{w}_i)) \right\}, \tag{37}$$

subject to capital, liquidity, legal, and exposure constraints. This is a mean-variance benchmark, not a complete description of sovereign or central-bank risk management [10].

9.3 Liquidity and solvency

Liquidity concerns the ability to meet obligations when due; solvency concerns the intertemporal value of assets, liabilities, and fiscal or institutional capacity. They must not be conflated. A stylized solvency condition is

$$\text{PV(future net resources)} \geq \text{current net liabilities}, \tag{38}$$

but public-sector solvency analysis must account for monetary institutions, contingent liabilities, taxation capacity, legal commitments, and the currency denomination of debt.

9.4 Financial asymmetry

The two entities may differ in leverage, liquidity, funding maturity, currency mismatch, collateral quality, market access, and risk-bearing capacity. A resource advantage can be offset by greater fragility. A robust comparison therefore evaluates both gross capacity and net risk exposure.

10 Information, Uncertainty, and Behaviour

10.1 Information structure

Let θ_i be a private type of entity i. A Bayesian game specifies type spaces, prior beliefs, and strategy functions. A strategy is then a mapping from information to actions rather than a fixed action. Incomplete information may concern costs, reserves, technology, preferences, policy intentions, or resilience.

10.2 Beliefs and expectations

Let $\mathcal{F}_t$ be the information available at time t. A rational expectation benchmark is

$$\mathbb{E}_t[Y_{t+1}] = \mathbb{E}[Y_{t+1} \mid \mathcal{F}_t]. \tag{39}$$

This is a consistency condition, not a claim that actual agents possess perfect information or compute exact conditional expectations.

10.3 Behavioural and psychological channels

Behavioural economics may introduce bounded rationality, loss aversion, reference dependence, ambiguity aversion, overconfidence, and strategic misperception [8, 15]. These concepts should be operationalized with measurable constructs and validated instruments. Psychiatric diagnoses must never be inferred from market behaviour, political decisions, or aggregate data. Clinical concepts are not substitutes for economic explanations.

10.4 Linguistic and narrative channels

Public statements, policy guidance, diplomatic language, and corporate communication can alter beliefs. Textual analysis may measure tone, uncertainty, commitment, and framing. Translation, sarcasm, institutional conventions, and semantic drift are major sources of measurement error. Language indicators should be validated against human-coded samples and out-of-sample performance.

11 Norway: Institutional and Economic Application

11.1 Norway as an application environment

Norway offers several distinct research environments: a sovereign fiscal system, a monetary authority, a financial system, energy and maritime industries, international trade relationships, and an institutional framework connected to the European Economic Area. These features make Norway suitable for several alternative duopoly studies. They do not, by themselves, prove that any particular Norwegian market or institution is a duopoly.

Statistics Norway's national accounts provide measures of GDP, consumption, investment, exports, imports, employment, wages, and industry-level activity [27]. The national accounts distinguish mainland economic activity from petroleum activities and ocean transport. This distinction is important when interpreting aggregate output, external exposure, and sectoral dependence.

11.2 Norges Bank

Norges Bank describes its mission as promoting economic stability and managing substantial assets on behalf of the Norwegian people. Its responsibilities include monetary policy, financial stability, payment systems, foreign exchange reserves, and management of the Government Pension Fund Global within the applicable mandate [22]. The Central Bank Act states the purposes of maintaining monetary stability, promoting financial system stability, and supporting an efficient and secure payment system; it also specifies a contribution to high and stable output and employment [23]. Monetary policy is

conducted by Norges Bank under the government-defined framework and the relevant legislation [24].

These institutional facts matter for model specification. A central bank cannot be modelled as an unconstrained profit-maximizing firm. Its mandate, governance, accountability, instruments, and statutory boundaries define its feasible action set.

11.3 Norwegian financial regulation

Finanstilsynet publishes information on laws and regulations governing banking and finance, including the Financial Institutions Act, anti-money laundering legislation, deposit guarantee arrangements, capital requirements, payment services, and lending practices [26]. The official Norwegian text is legally authoritative; English translations are provided for information and should not replace legal verification.

Norges Bank's account of Norway's financial system explains that the EEA Agreement shapes the framework conditions for the Norwegian financial system and connects Norway to the EU single market for financial services [25]. An empirical study must identify the relevant legal version, implementation date, and institutional competence rather than treating "European regulation" as a single undifferentiated variable.

11.4 Possible Norway research designs

Table 3: Potential duopoly applications involving Norway

Application	Candidate entities	Main identification issue
Sovereign interaction	Norway and a specified foreign state	Separate bilateral effects from common global shocks
Monetary interaction	Norges Bank and a specified foreign central bank	Identify policy transmission and exchange-rate channels
Currency competition	NOK and a specified currency	Define the relevant function: reserve, settlement, invoicing, or store of value
Banking	Two specified banks active in Norway	Establish relevant product and geographic markets
Energy	Two specified producers or blocs	Account for regulation, capacity, contracts, and global prices
Payment infrastructure	Two specified networks or providers	Measure substitution, interoperability, and network effects
Technology	Two specified platforms or standards	Define user-side and developer-side market boundaries

The table lists research designs, not findings. The selected pair must be supported by evidence that the entities are strategically relevant in the specified domain.

11.5 Norway-specific empirical discipline

A credible Norwegian application should:

1. define the entity pair and the analytical domain;
2. specify the relevant legal and institutional period;
3. establish market boundaries or the relevant bilateral relationship;
4. identify available data and measurement limitations;
5. distinguish domestic, mainland, petroleum, and external channels where relevant;
6. control for global commodity prices, interest rates, exchange rates, and common shocks;
7. report uncertainty and robustness checks;
8. avoid causal language when the design identifies only association.

12 Law, Regulation, and Institutional Design

12.1 Legal feasibility

The action set of an entity is legally constrained. Let $\mathcal{X}_i^0$ be its technologically and economically feasible actions, and let $\mathcal{L}_i$ be the set permitted by applicable law. Then

$$\mathcal{X}_i = \mathcal{X}_i^0 \cap \mathcal{L}_i. \tag{40}$$

This formulation is schematic: legal rules may impose duties, procedures, disclosure requirements, or standards that cannot be represented by a simple set restriction.

12.2 Competition law

Competition analysis begins with market definition, competitive constraints, market power, conduct, and effects. A two-firm or two-entity count does not automatically establish dominance or unlawful conduct. Nor does parallel behaviour alone prove an agreement. For Norway, the applicable national competition rules and relevant EEA provisions must be examined in their current legal form. This manuscript does not provide a legal opinion.

12.3 Public institutions

Public authorities may have statutory objectives that differ from private utility maximization. The model should represent legal mandates as constraints, objective components, or both. It should also distinguish operational independence, political accountability, ownership, and legal personality. Norges Bank's separate legal personality and state ownership illustrate why these concepts should not be collapsed into a single variable [23, 22].

12.4 Ethics and rights

A welfare-maximizing equilibrium is not necessarily lawful or ethically acceptable. Rights, due process, privacy, non-discrimination, democratic accountability, and distributional justice may operate as binding constraints rather than as monetized terms in a scalar objective function. Ethical evaluation should state its normative framework and avoid presenting contested moral premises as mathematical facts.

13 Welfare Economics and Distribution

13.1 Social welfare

Let W be a social welfare function:

$$W(\mathbf{x}) = \sum_{h=1}^{H} \omega_h U_h(\mathbf{x}), \tag{41}$$

where h indexes affected households or groups and ω_h are explicit welfare weights. In a sovereign application, welfare may include income, employment, health, security, environmental quality, resilience, and intergenerational outcomes.

13.2 Surplus and externalities

For a conventional product market, total surplus is the sum of consumer and producer surplus, adjusted for external costs and benefits. In a broader institutional duopoly, welfare accounting must include spillovers, systemic risk, public goods, distribution, and non-market outcomes. A Nash equilibrium need not maximize W.

13.3 Distributional incidence

Let Δy_h be the change in income or welfare for group h. An aggregate positive change can coexist with losses for some groups. The analysis should therefore report distributional effects, not only aggregate surplus.

14 Geopolitics, Diplomacy, and Security Economics

A sovereign duopoly requires a broader objective specification than a commercial market. Relevant dimensions may include national welfare, security, economic resilience, strategic autonomy, trade access, and international commitments. These objectives can conflict.

A schematic sovereign objective is

$$U_i = \alpha_i W_i + \beta_i S_i + \gamma_i R_i + \delta_i D_i, \tag{42}$$

where W_i is domestic welfare, S_i security, R_i resilience, and D_i diplomatic or institutional value. The weights are modelling parameters, not universal constants. They should be justified and subjected to sensitivity analysis.

Diplomacy may change the feasible set by creating treaties, information channels, verification procedures, or dispute-resolution mechanisms. Repeated interaction and credible commitments can alter strategic incentives. International law and treaty obligations

should be modelled as constraints only after their applicability and legal status are established.

14.1 Warfare economics and conflict risk

Security competition may involve deterrence, arms expenditure, sanctions, supply-chain restrictions, and escalation risk. A responsible economic analysis should model conflict as a costly risk and should not translate abstract game theory into operational instructions for violence. The relevant outcomes include expected civilian harm, economic disruption, fiscal costs, refugee flows, infrastructure damage, and the probability of escalation.

A stylized expected social cost is

$$\mathbb{E}[C_{\text{conflict}}] = \sum_{s \in \mathcal{S}} \Pr(s) C(s), \tag{43}$$

where scenarios s include peaceful, coercive, crisis, and conflict states. The probabilities and costs require transparent evidence and uncertainty ranges.

15 Physics, Chemistry, Biology, and Environmental Constraints

The use of concepts from the natural sciences must be disciplined. Physical laws are not metaphors for market behaviour; they impose real constraints on production, energy, transport, infrastructure, and environmental impact.

15.1 Physics and energy

Energy balance, capacity, thermodynamic efficiency, network congestion, and storage constraints may determine feasible economic actions. A simple energy constraint is

$$E_{\text{input}} = E_{\text{useful}} + E_{\text{loss}}, \tag{44}$$

with efficiency

$$\eta = \frac{E_{\text{useful}}}{E_{\text{input}}}, \qquad 0 \leq \eta \leq 1. \tag{45}$$

For energy-sector applications involving Norway, physical capacity, reservoir conditions, grid constraints, pipeline infrastructure, and environmental limits must be measured rather than inferred from financial data alone.

15.2 Chemistry and materials

Chemical composition, material quality, corrosion, storage stability, and process safety can constrain production and infrastructure. A model of two competing producers should include material input requirements and process constraints where these materially affect capacity or cost.

15.3 Biology and public health

Biological systems may enter through labour productivity, demographic change, public health, fisheries, ecosystems, and environmental externalities. Health and ecological outcomes should be measured with appropriate scientific indicators. They should not be reduced to arbitrary monetary proxies where such reduction would conceal material effects.

15.4 Environmental accounting

Let e_i denote emissions or environmental pressure attributable to entity i. A social objective may include

$$W_{\text{adjusted}} = W_{\text{economic}} - \lambda_E(e_1 + e_2) - \lambda_B B_{\text{biodiversity}}, \tag{46}$$

where $B_{\text{biodiversity}}$ represents a defined biodiversity loss measure. The weights require normative and empirical justification.

16 Econometric Framework

16.1 Research estimand

Before estimation, define the estimand. Examples include the effect of entity 1's policy on entity 2's outcome, the effect of strategic entry on concentration, or the response of Norwegian financial conditions to an external policy shock. The estimand must identify the population, treatment, outcome, horizon, and counterfactual.

16.2 Panel model

A generic panel specification is

$$y_{it} = \alpha_i + \lambda_t + \beta x_{it} + \theta x_{jt} + \boldsymbol{\gamma}'\mathbf{z}_{it} + \varepsilon_{it}, \qquad i \neq j, \tag{47}$$

where α_i and λ_t are entity and time effects. The cross-entity coefficient θ is not automatically causal. Simultaneity, common shocks, omitted variables, measurement error, and reflection problems must be addressed [9, 1, 2].

16.3 Simultaneity

If both entities respond to one another contemporaneously, ordinary least squares may be inconsistent. A structural system may be written

$$y_{1t} = \alpha_1 + \beta_{12} y_{2t} + \boldsymbol{\gamma}_1'\mathbf{z}_{1t} + u_{1t}, \tag{48}$$

$$y_{2t} = \alpha_2 + \beta_{21} y_{1t} + \boldsymbol{\gamma}_2'\mathbf{z}_{2t} + u_{2t}. \tag{49}$$

Identification requires valid exclusion restrictions, instruments, timing assumptions, or an alternative design. A high correlation between y_1 and y_2 does not identify β_{12} or β_{21}.

16.4 Time-series dynamics

For Norwegian macro-financial data, possible tools include vector autoregressions, local projections, cointegration analysis, error-correction models, and structural time-series models. A VAR is

$$\mathbf{y}_t = \mathbf{c} + \sum_{k=1}^{p} \mathbf{A}_k \mathbf{y}_{t-k} + \mathbf{u}_t. \tag{50}$$

Impulse responses depend on the identification of structural shocks. Reduced-form innovations should not automatically be labelled policy shocks [16, 18, 7].

16.5 Causal designs

Potential designs include difference-in-differences, event studies, instrumental variables, regression discontinuity, synthetic controls, and structural estimation. Each design requires assumptions such as parallel trends, exclusion restrictions, continuity, no anticipation, or correct structural specification. These assumptions must be argued and tested where possible.

16.6 Concentration measures

For market shares s_i, the Herfindahl–Hirschman index is

$$HHI = \sum_{i=1}^{N} s_i^2, \qquad \sum_{i=1}^{N} s_i = 1. \tag{51}$$

For a market containing exactly two firms with shares s and $1 - s$,

$$HHI = s^2 + (1 - s)^2. \tag{52}$$

This measure is meaningful only after defining the relevant market and measuring shares. It is not a general measure of strategic power for every sovereign or institutional dyad.

17 Statistics, Uncertainty, and Robustness

17.1 Sampling uncertainty

An estimator $\widehat{\theta}$ should be accompanied by a standard error, confidence interval, or posterior distribution as appropriate. Statistical significance does not establish economic significance, causal validity, or policy desirability.

17.2 Model uncertainty

Alternative specifications should be compared. Sensitivity analysis should vary functional forms, sample periods, lags, market definitions, priors, instrument sets, and treatment definitions. Multiple testing and researcher degrees of freedom should be disclosed.

17.3 Stress testing

A stress test evaluates outcomes under specified adverse scenarios. It is not a forecast unless probabilities and predictive calibration are supplied. Scenarios for a Norway application may include commodity-price shocks, exchange-rate volatility, global funding stress, cyber incidents, supply-chain disruption, or regulatory changes.

18 Computer Science, Algorithms, and AI/ML

18.1 Computational representation

The model can be represented as a typed data structure containing:

1. entity identifiers and institutional metadata;
2. action spaces and constraints;
3. objective-function parameters;
4. state variables and observations;
5. interaction channels and edge weights;
6. legal and temporal regimes;
7. model version, data provenance, and uncertainty.

18.2 Equilibrium computation

For differentiable concave games, a numerical solver may seek a root of the stacked first-order conditions. For general games, algorithms may use best-response iteration, complementarity solvers, fixed-point methods, or mixed-strategy linear programming. Convergence is not guaranteed without appropriate assumptions.

18.3 Illustrative pseudocode

```
INPUT:
entity definitions A1, A2
domain Omega
feasible action sets X1, X2
objective functions U1, U2
tolerances epsilon and maximum iterations M

INITIALIZE:
choose feasible actions x1, x2
set iteration k = 0

REPEAT:
x1_new = best_response_1(x2)
x2_new = best_response_2(x1_new)
```

```
residual_1 = distance(x1_new, x1)
residual_2 = distance(x2_new, x2)

x1 = x1_new
x2 = x2_new
k = k + 1

UNTIL:
max(residual_1, residual_2) < epsilon
OR k = M

VERIFY:
test unilateral deviations
check feasibility and constraints
report convergence status and residuals

OUTPUT:
candidate equilibrium, diagnostics, and sensitivity results
```

The algorithm is a computational template. Best-response iteration may cycle or diverge; a failure to converge is a result to diagnose, not a reason to report a false equilibrium.

18.4 Machine learning

Supervised learning may forecast outcomes or classify regimes; unsupervised learning may identify clusters; reinforcement learning may model adaptive policies; and multi-agent reinforcement learning may simulate strategic adaptation. Predictive performance does not establish causal structure. A learned policy must be evaluated for robustness, interpretability, distribution shift, and constraint compliance.

18.5 Cybersecurity and data governance

A computational duopoly model may be exposed to data poisoning, model extraction, adversarial inputs, unauthorized access, and supply-chain risks. Controls should include access management, encryption in transit and at rest, audit logs, reproducible environments, dependency review, backups, and incident response. Sensitive personal or institutional data should be minimized and processed under applicable law. Cybersecurity testing should be defensive, authorized, and proportionate.

19 The Duopoly Transition Problem

19.1 From monopoly to duopoly

Entry may occur when expected post-entry returns exceed entry costs:

$$\mathbb{E}[\Pi_{\text{entrant}}] - F_{\text{entry}} \geq 0. \tag{53}$$

This is a simplified condition. Entry also depends on finance, regulation, technology, network effects, strategic deterrence, and uncertainty.

19.2 From duopoly to oligopoly

A duopoly may become an oligopoly if additional entities become strategically relevant. Let $N_{\text{eff}}(t)$ denote the number of effective strategic participants under a declared rule. A transition occurs when

$$N_{\text{eff}}(t) > 2. \tag{54}$$

The threshold is a classification rule, not a theory of market power. The effective-participant measure must be defined and validated.

19.3 From duopoly to monopoly

A transition toward monopoly may occur through exit, merger, acquisition, network tipping, innovation, regulatory privilege, or resource asymmetry. A single dominant participant does not necessarily have an unconstrained monopoly if imports, potential entry, substitutes, or regulation impose effective constraints.

19.4 Reversibility

The transitions are not necessarily irreversible. New technology, legal intervention, interoperability, entry, or changes in demand may reverse concentration. The model should allow multiple transition paths.

20 Propositions and Research Hypotheses

The following propositions are research targets. Their validity depends on the assumptions and evidence specified in each application.

Proposition 20.1 (Institutional generality)**.** *If two entities have well-defined action spaces, constraints, interaction rules, and objective or behavioural criteria, then the formal analysis of their strategic interaction can be conducted independently of their institutional labels.*

Proof. A strategic game is specified by players, strategy spaces, and payoff or preference relations. The equilibrium definition refers to feasible unilateral deviations and the resulting preference ordering. It does not require that the players be firms. Institutional identity matters through the construction of the feasible sets, preferences, information, and legal constraints. Therefore, preserving these structural objects preserves the formal game-theoretic analysis. □

Proposition 20.2 (Asymmetry does not preclude equilibrium)**.** *Unequal resources or objectives do not, by themselves, preclude the existence of a Nash equilibrium.*

Proof. The standard equilibrium-existence conditions do not require equal strategy sets, identical payoff functions, or equal resource endowments. They require appropriate properties of each player's own strategy set and payoff function. Consequently, asymmetry is compatible with equilibrium whenever the relevant existence conditions hold. □

Proposition 20.3 (Equilibrium does not imply welfare optimality)**.** *A Nash equilibrium need not maximize a social welfare function.*

Proof. Nash equilibrium rules out profitable unilateral deviations by individual players. A social welfare maximum compares joint outcomes and may require coordinated changes. These are different optimization problems. Standard externality and market-power examples demonstrate that equilibrium can be socially inefficient. □

Hypothesis 20.4 (Competitive asymmetry). Conditional on market definition and controls, differences in finance, technology, information, and network reach may predict differences in strategic influence and persistence.

Hypothesis 20.5 (Institutional concentration). In some settings, resource and information asymmetries may increase the probability that decision-making becomes concentrated. This hypothesis requires an operational measure of institutional concentration and a causal design.

Hypothesis 20.6 (Endogenous entry). Innovation, interoperability, changes in regulation, and access to capital may alter the number of strategically relevant entities in a domain.

Hypothesis 20.7 (Dynamic instability). A duopoly may exhibit regime shifts, cycles, or loss of stability when strategic feedback, shocks, or state constraints change.

21 Philosophical and Methodological Foundations

The theory is realist about institutional constraints but pluralist about objectives. It treats mathematical models as conditional representations, not as complete descriptions of social reality. The choice of entities, boundaries, variables, and welfare weights is itself a substantive decision.

A model is useful when it clarifies mechanisms, exposes assumptions, generates testable implications, and survives comparison with evidence. Formal elegance does not guarantee empirical relevance. Nor does predictive success alone establish a causal explanation.

Religious and philosophical traditions may inform ethical reflection on justice, stewardship, obligation, peace, and intergenerational responsibility. They should be represented transparently as normative frameworks rather than misstated as empirical or mathematical laws. Political legitimacy and institutional accountability likewise cannot be inferred from equilibrium conditions alone.

22 Empirical Research Protocol for Norway

22.1 Step 1: Select the dyad

Name A_1 and A_2, define their legal identities, and specify the domain Ω. State why the pair, rather than a larger set, is the appropriate focal system.

22.2 Step 2: Define the outcome

Specify the dependent variable, unit of observation, frequency, sample period, and economic interpretation. Examples include policy rates, lending spreads, market shares, payment volumes, bilateral trade, investment flows, or network centrality.

22.3 Step 3: Map the institutional environment

Record relevant statutes, regulations, mandates, ownership, governance, treaties, market rules, and implementation dates. Distinguish legal text from administrative practice.

22.4 Step 4: Assemble data

Potential sources include Statistics Norway national accounts, Norges Bank statistics and publications, Finanstilsynet material, relevant market operators, international organizations, and legally accessible company disclosures. The data dictionary must record source, definition, unit, frequency, revisions, missingness, and access conditions.

22.5 Step 5: Estimate and validate

Estimate descriptive, structural, and causal models separately. Conduct diagnostics, robustness checks, placebo tests, out-of-sample validation, and sensitivity analysis. Publish code and non-confidential data transformations where legally and contractually possible.

22.6 Step 6: Interpret

Separate statistical findings, economic mechanisms, legal conclusions, and normative judgments. State what the evidence does not establish.

23 Limitations

The general framework is intentionally broad. That breadth creates risks. First, an overly broad definition may classify ordinary bilateral influence as a duopoly. The domain and strategic relevance criteria are therefore essential. Second, objective functions may be difficult to observe, especially for public institutions and sovereign states. Third, legal constraints are context-specific and may change over time. Fourth, strategic interaction creates simultaneity and identification problems. Fifth, network effects and external participants may make a two-player abstraction incomplete. Sixth, machine-learning models may predict patterns without identifying mechanisms.

The Norway application cannot be completed empirically until a specific pair of entities, a domain, a sample period, and data are selected. No conclusion about a Norwegian duopoly is asserted merely from the existence of Norwegian institutions or markets.

24 Conclusion

A duopoly is best understood as a domain-specific system of two strategically relevant entities, not as a synonym for a market containing two firms. The entities may be homogeneous or heterogeneous, symmetric or asymmetric, competitive or partly cooperative, domestic or international, private or public. The formal analysis requires clear action spaces, constraints, objectives, information structures, and interaction channels.

The resulting framework connects classical industrial organization with game theory, dynamic systems, finance, econometrics, network science, institutional analysis, law, public policy, computational modelling, and security economics. Its Norway application

must be selected and identified with care. Norges Bank's statutory role, Norway's financial regulatory framework, the EEA context, and the national accounts provide important institutional foundations, but they do not predetermine the dyad or its equilibrium.

The principal conclusion is methodological: the identity of the participants is an empirical and institutional question; the structure of their interdependence is the mathematical object of analysis.

A Notation

Symbol	Meaning
$\mathcal{A}$	Set of two focal entities
A_i	Entity i
Ω	Analytical domain
$\mathcal{E}$	External environment
$\mathcal{S}$	State space
$\mathcal{X}_i$	Feasible action set of entity i
x_i	Action or strategy of entity i
U_i	Objective or preference representation of entity i
BR_i	Best-response correspondence
$\mathbf{Z}$	System state vector
$\mathbf{J}_F$	Jacobian of the state-transition system
$\mathbf{R}_i$	Resource vector of entity i
K_i	Capital stock or financial resources
HHI	Herfindahl–Hirschman concentration index
N_{eff}	Number of effective strategic participants
ρ_i	Discount rate of entity i
ε_t	Shock or innovation
$\mathcal{F}_t$	Information set at time t

References

[1] J. D. Angrist and J.-S. Pischke, *Mostly Harmless Econometrics: An Empiricist's Companion*, Princeton University Press, Princeton, 2009.

[2] M. Arellano and S. Bond, Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations, *Review of Economic Studies*, 58(2), 277–297, 1991.

[3] J. Bertrand, Théorie mathématique de la richesse sociale, *Journal des Savants*, 1883, 499–508.

[4] A. A. Cournot, *Researches into the Mathematical Principles of the Theory of Wealth*, original French edition, 1838; English translation, Macmillan, 1897.

[5] D. Fudenberg and J. Tirole, *Game Theory*, MIT Press, Cambridge, Massachusetts, 1991.

[6] I. L. Glicksberg, A further generalization of the Kakutani fixed point theorem, with application to Nash equilibrium points, *Proceedings of the American Mathematical Society*, 3, 170–174, 1952.

[7] Ò. Jordà, Estimation and inference of impulse responses by local projections, *American Economic Review*, 95(1), 161–182, 2005.

[8] D. Kahneman and A. Tversky, Prospect theory: An analysis of decision under risk, *Econometrica*, 47(2), 263–291, 1979.

[9] C. F. Manski, Identification of endogenous social effects: The reflection problem, *Review of Economic Studies*, 60(3), 531–542, 1993.

[10] H. Markowitz, Portfolio selection, *Journal of Finance*, 7(1), 77–91, 1952.

[11] A. Mas-Colell, M. D. Whinston, and J. R. Green, *Microeconomic Theory*, Oxford University Press, New York, 1995.

[12] J. F. Nash, Equilibrium points in n-person games, *Proceedings of the National Academy of Sciences*, 36(1), 48–49, 1950.

[13] J. F. Nash, Non-cooperative games, *Annals of Mathematics*, 54(2), 286–295, 1951.

[14] J. B. Rosen, Existence and uniqueness of equilibrium points for concave N-person games, *Econometrica*, 33(3), 520–534, 1965.

[15] H. A. Simon, A behavioral model of rational choice, *Quarterly Journal of Economics*, 69(1), 99–118, 1955.

[16] C. A. Sims, Macroeconomics and reality, *Econometrica*, 48(1), 1–48, 1980.

[17] H. von Stackelberg, *Marktform und Gleichgewicht*, Springer, Vienna, 1934.

[18] J. H. Stock and M. W. Watson, Vector autoregressions, *Journal of Economic Perspectives*, 15(4), 101–115, 2001.

[19] P. D. Taylor and L. B. Jonker, Evolutionarily stable strategies and game dynamics, *Mathematical Biosciences*, 40, 145–156, 1978.

[20] J. Tirole, *The Theory of Industrial Organization*, MIT Press, Cambridge, Massachusetts, 1988.

[21] J. W. Weibull, *Evolutionary Game Theory*, MIT Press, Cambridge, Massachusetts, 1995.

[22] Norges Bank, Our mission and core responsibilities, official institutional description, accessed 4 November 2026.

[23] Norges Bank, Act relating to Norges Bank and the Monetary System, etc. (Central Bank Act), official English-language information and statutory text, accessed 4 November 2026. The Norwegian legal text is authoritative.

[24] Norges Bank, FAQ: Monetary policy, official institutional explanation, accessed 4 November 2026.

[25] Norges Bank, *Norway's Financial System 2024*, official report and web publication, 2024.

[26] Finanstilsynet, Banking and finance: Laws and regulations, official regulatory information, accessed 4 November 2026.

[27] Statistics Norway, National accounts and business cycles, official national accounts statistics and documentation, accessed 4 November 2026.

Glossary

Action space
The set of actions available to an entity under its technological, economic, legal, and institutional constraints.

Asymmetric duopoly
A two-entity strategic system in which relevant resources, objectives, information, constraints, or influence differ.

Best response
An action that maximizes an entity's objective, given the action of the other entity and the model's constraints.

Duopoly
A domain-specific strategic system with two strategically relevant focal entities.

Dynamic game
A strategic interaction in which states and actions evolve over time and current choices affect future opportunities or outcomes.

Effective strategic participant
An entity whose actions materially constrain or alter the strategic outcomes in the specified domain.

Equilibrium
A strategy profile satisfying a specified solution concept, such as Nash equilibrium.

Externality
A cost or benefit imposed on parties not fully represented in the decision-maker's private objective.

Institutional constraint
A legal, governance, procedural, or mandate-based restriction on an entity's feasible actions.

Market definition
The analytical identification of products, services, and geographic areas within which competitive constraints are assessed.

Nash equilibrium
A strategy profile in which no player can improve its objective through a unilateral feasible deviation.

Nested duopoly
: A two-entity system at one level of analysis whose constituent entities contain more complex structures internally.

Strategic interdependence
: A relationship in which an entity's choices affect the other entity's incentives, constraints, or attainable outcomes.

Structural identification
: The use of assumptions, instruments, timing, or experimental variation to recover parameters of an economic mechanism.

Systemic risk
: The risk that distress or disruption in one part of a system propagates and impairs the functioning of the wider system.

The End